GlassQuill

Explainable detection of AI-generated text

Applications and Use Cases of Machine Learning (TE7483) · AIML, SIT Pune
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Agenda

12 minutes, then questions and a live demo

01 **Problem and use case**
why explainable detection, who needs it, scope
2 min

02 **Data and preprocessing**
4 public datasets, cleaning, 18 features
3 min

03 **Methodology**
pipeline, four models, verdict rule, validation
2 min

04 **Implementation**
code structure, reproducibility, the web app
2 min

05 **Results and analysis**
metrics, comparison, robustness, explanations
2 min

06 **Conclusion**
SWOT, findings and future scope
1 min

01

The problem

Colleges, newsrooms and hiring teams use AI-text detectors, but most return a bare score. A student who is wrongly flagged has no way to question it.

Liang et al. (2023) found popular detectors flagged over half of essays by non-native English writers as AI. GlassQuill labels English text as human or AI, explains why, and treats a low false positive rate as the main goal.

Research questions

Build an explainable human-vs-AI classifier for English text (50 to 400 words), keep humans wrongly flagged as AI low, and test it on unseen generators and paraphrased text.

01.

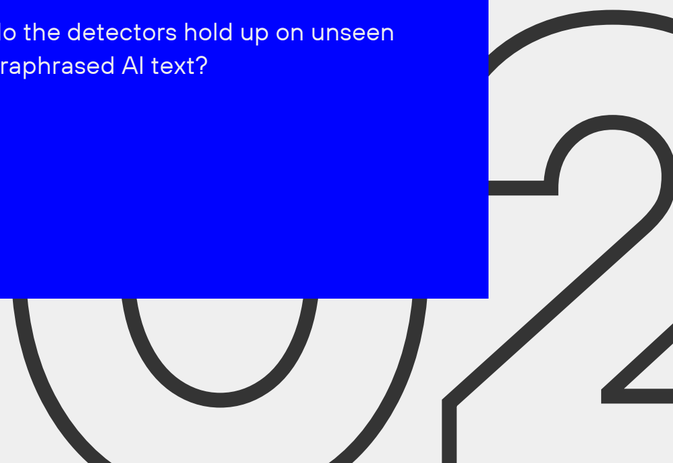
RQ1 Which simple signals (perplexity, sentence rhythm, vocabulary, punctuation) separate human and AI text?

02.

RQ2 How close can explainable models (XGBoost + SHAP) get to a black box (DistilBERT)?

03.

RQ3 How well do the detectors hold up on unseen LLMs and on paraphrased AI text?



Use case, scope and refinement

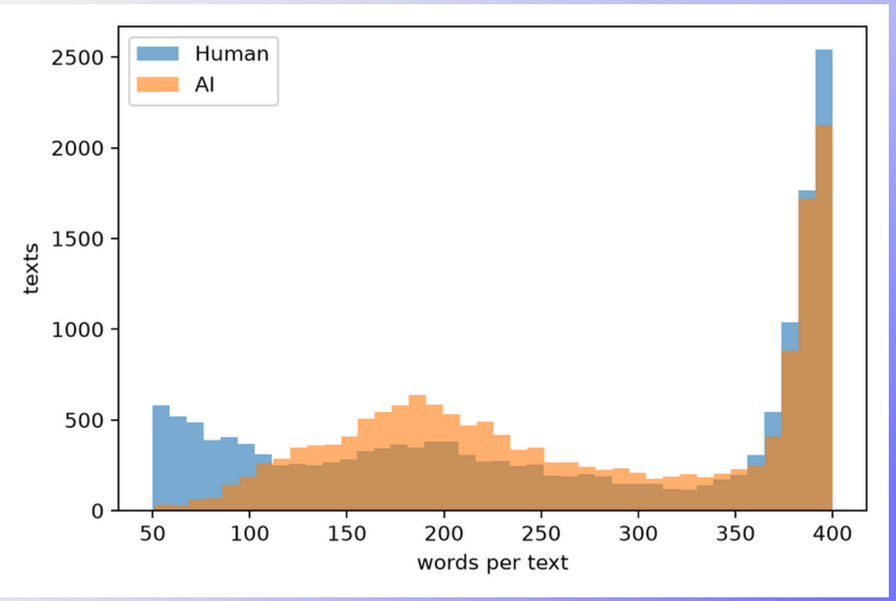
Domain: education and content moderation · Task: binary NLP text classification (0 = human, 1 = AI)

Stakeholder	What they need
Teachers, universities	a second opinion with reasons, not a verdict
Students, writers	a fair way to question a flag
Platforms, newsrooms	screening at scale with an audit trail
Researchers	a transparent baseline for detectors

Phase I plan	Final project
HC3 only (ChatGPT 2022)	+ HAP-E and M4GT: 2024 models, 17 domains
Raw HC3 text	shortcut cleaning (99.3% was fake accuracy)
XGBoost as the verdict	mean of XGBoost + LR: FPR 6.6% to 2.0%
Gradio demo in Colab	React + FastAPI app, 28 tests, script pipeline

In scope: English text, 50 to 400 words, many genres (Q&A, essays, news, fiction, academic, Reddit, Wikipedia).
Out of scope: other languages, source code, texts that mix human and AI writing, naming which LLM wrote a text.

Datasets



HC3 (ChatGPT vs human answers to the same questions), **HAP-E** (GPT-4o, Llama-3; 6 genres) and **M4GT** (GPT-4, Mixtral, Gemma, Cohere): 31,984 texts in 17 domains, 50% human and 50% AI in each.

Target: label (0 = human, 1 = AI), no missing values after cleaning. Split 70/15/15 by question or document. **MAGE** (27 generators, 2,000 texts) is held out as the out-of-domain test.

03

Attributes, target and validation

One row per text after cleaning: 31,984 texts, 0 missing values

Column	Type	Meaning
text	string	the passage: model input
label	int 0/1	target: 0 = human, 1 = AI
qid	string	question / document id, for splitting
source	category	one of 17 domains
generator	category	human or the model name
corpus	category	HC3, HAP-E or M4GT
n_words	int	words after cleaning (50 to 400)

Split	Texts	Share AI
Train (70%)	22,371	0.500
Validation (15%)	4,784	0.500
Test (15%)	4,829	0.501
MAGE (unseen)	2,000	0.500

Missing values: empty and duplicate texts dropped; features divide by max(words, 1), so none are missing.
Validation: GroupShuffleSplit by question or document, 5-fold GroupKFold for tuning, test set used once.

04

Data cleaning

01

Tokenizer leftovers: 34% of HC3 human texts had “do n't” vs 0.02% of AI texts, plus “(x)” and URL_0.

02

Balanced per domain: open_qa had 20 human vs 764 AI texts, so the topic predicted the label.

03

Markdown bold and headings in AI text, curly quotes and literal “\n” were normalised for every source.

04

Split by question or document and tuned with GroupKFold, so no topic is in both train and test.

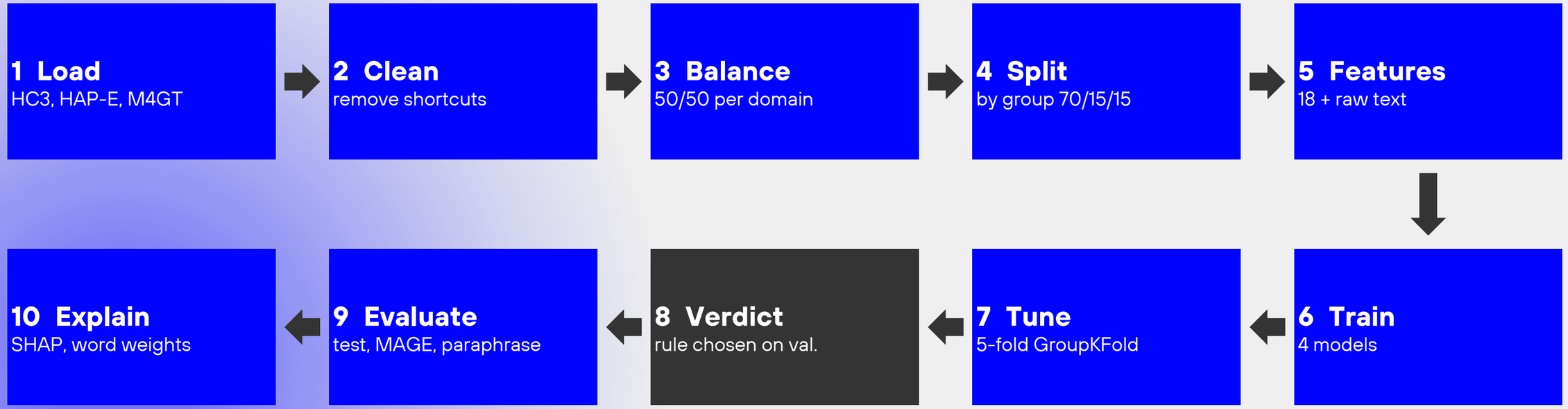
The 18 features

6 from GPT-2: how predictable each word is (log perplexity, its spread and maximum, top-1 and top-10 guess rates, entropy). AI text is easier to predict: perplexity is the strongest signal (-1.4 SD).

12 on writing style: sentence length and burstiness (humans 0.53 vs AI 0.38), vocabulary richness, word length, punctuation, first-person words, contractions and readability.

The ML pipeline

run_pipeline.py: eight cached stages, seed 42, reproducible with one command



Every stage saves its output, so any stage can be rerun alone. python run_pipeline.py --quick reproduces all stages on a sample in about 10 minutes.

Four models

and one verdict

01

Perplexity threshold on GPT-2 log perplexity alone; cut-off chosen on validation. Zero-shot baseline.

02

TF-IDF + logistic regression on 1-2 word features; C tuned by 5-fold GroupKFold. Glass box on words.

03

XGBoost on the 18 features, monotone constraints where theory and data agree. Explained with SHAP.

04

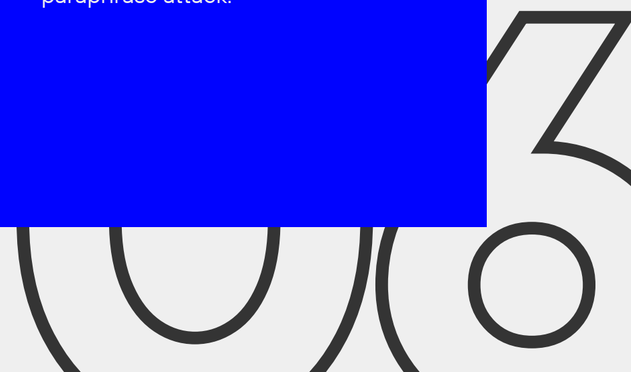
DistilBERT fine-tuned for 2 epochs. Most powerful, but a black box: kept for comparison.

05

Verdict = mean of XGBoost and TF-IDF + LR, chosen on validation. Unclear band: 35 to 65%.

06

Evaluation: test set used once, MAGE for unseen generators, a T5 paraphrase attack.

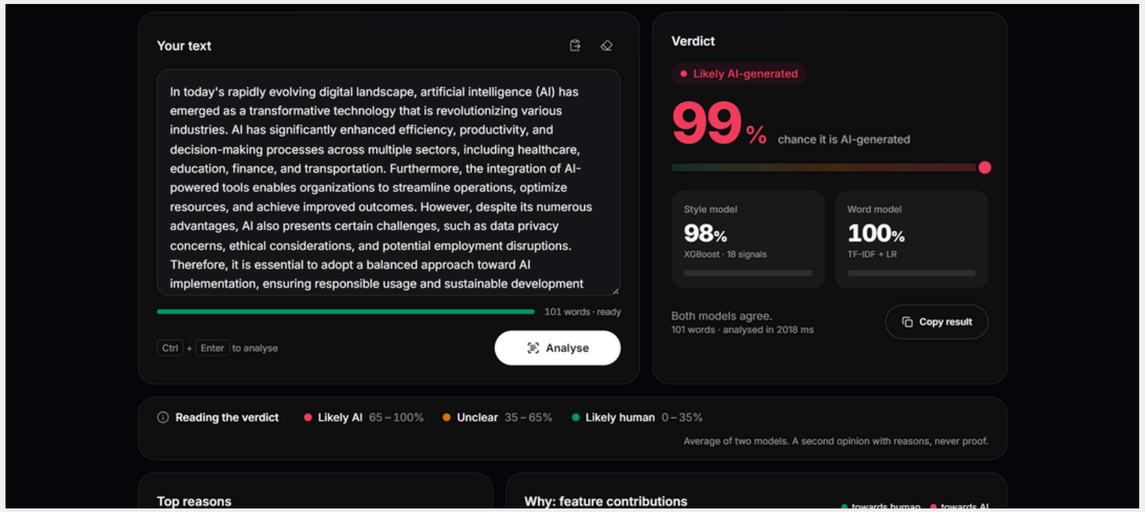


Implementation

Complete, tested and reproducible: code, pipeline and a working web app

Layer	Tools	Code
Data, features	pandas, GPT-2, textstat	src/data.py, features.py
Models	scikit-learn, XGBoost, DistilBERT	src/models.py
Explanations	SHAP, LIME	src/explain.py
Pipeline	8 cached stages, seed 42	run_pipeline.py
Web app	FastAPI + React	server.py, web/
Quality	28 tests, lint, types	tests/

Python 3.11, PyTorch 2.11 on an RTX 3050 laptop GPU. Full run about 1.5 h; the app answers in about 0.1 s per text.



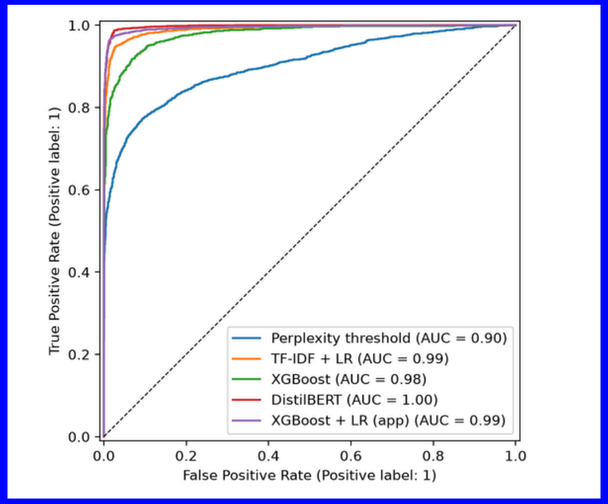
The detector page: verdict, both model scores and the evidence below it.

97.4%

Test accuracy

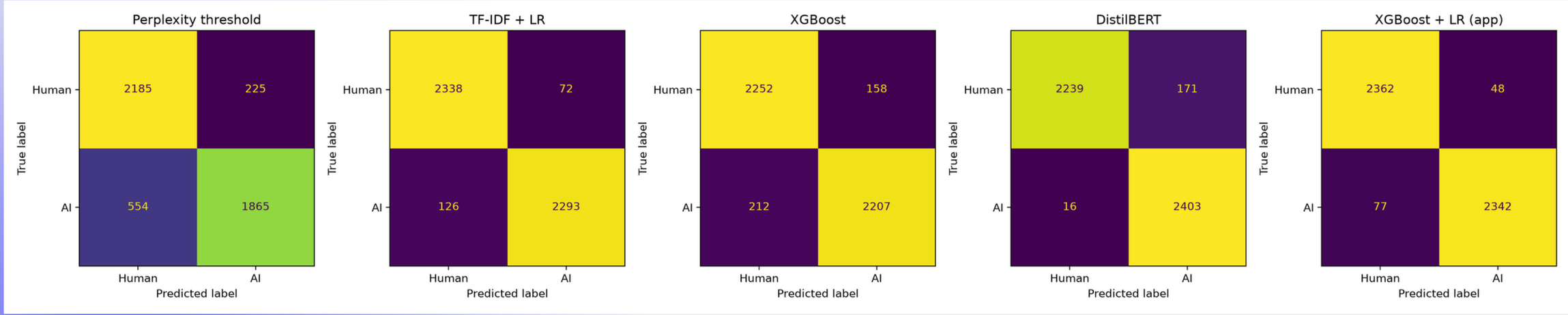
On 4,829 unseen test texts the verdict beats every single model and wrongly flags only 2.0% of human texts.

Model	Acc.	Prec.	Recall	F1	FPR
GlassQuill verdict	**97.4%**	**0.980**	**0.968**	**0.974**	**2.0%**
DistilBERT	96.1%	0.934	0.993	0.963	7.1%
TF-IDF + LR	95.9%	0.970	0.948	0.959	3.0%
XGBoost	92.3%	0.933	0.912	0.923	6.6%
Perplexity threshold	83.9%	0.892	0.771	0.827	9.3%



Comparative analysis

Same metrics for every model, in-domain, on unseen generators and under attack



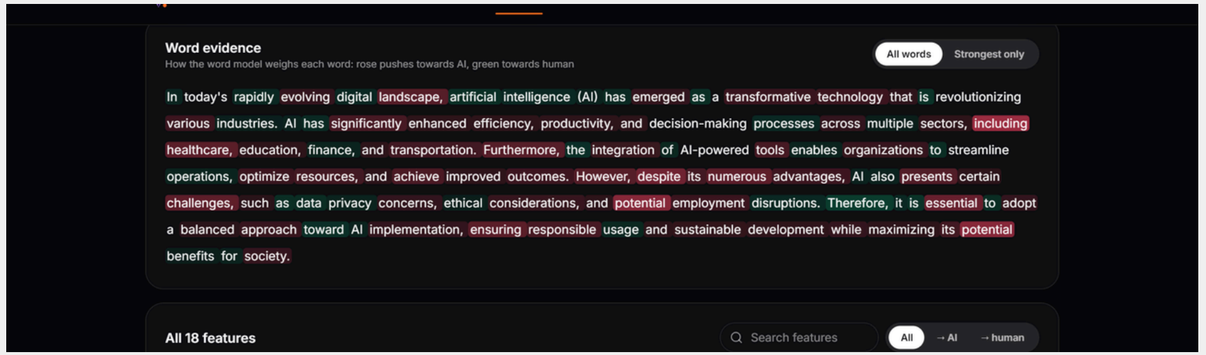
MAGE (27 unseen generators)	Acc.	F1	FPR
GlassQuill verdict	**66.4%**	**0.542**	**6.8%**
Perplexity threshold	65.6%	0.551	11.1%
DistilBERT	65.5%	0.585	17.6%
XGBoost	63.4%	0.532	14.9%
TF-IDF + LR	60.6%	0.426	7.9%

Paraphrase attack (recall)	Before	After
GlassQuill verdict	**98%**	**93%**
DistilBERT	100%	100%
XGBoost	100%	95%
Perplexity threshold	100%	93%
TF-IDF + LR	93%	77%

Showing the evidence

SHAP shows how much each of the 18 signals moved the style model; low perplexity pushes hardest towards AI. The word model's weights are exact, so every word is coloured towards human or AI.

The same view, live in the app:



09

SWOT

Weaknesses

66.4% on 27 unseen MAGE generators. Small open models (OPT, BLOOM, GPT-J) go undetected.

Threats

Paraphrasing (98% to 93% caught), plain writing by non-native writers, and new LLMs.

Strengths

97.4% accuracy and 2.0% false positives on the test set. Every verdict comes with reasons.

Opportunities

A modern reader model or Binoculars, sentence-level detection, calibrated probabilities.

10

Conclusion and next steps

Two explainable models, averaged, beat a fine-tuned DistilBERT (97.4% vs 96.1%) with under a third of its false positives, and every verdict can be checked. Removing dataset shortcuts mattered most.

Next: a threshold for a fixed 1% FPR, sentence-level detection for mixed texts, the RAID benchmark and Indian languages. A detector score is a second opinion, never proof.

Thank You

Questions?